

Induced polarization of shaly sands

H. J. Vinegar* and M. H. Waxman*

ABSTRACT

Equations have been developed that relate induced polarization (IP) in shaly sands to measurable petrophysical parameters. The induced-polarization process has been modeled in terms of two mechanisms: clay counterion displacement and membrane blockage. The resulting equations can be used to determine shaliness, brine conductivity, and oil saturation from in-phase and out-of-phase conductivities. Laboratory measurements have confirmed the IP dependence on these variables, as well as on temperature.

INTRODUCTION

Electrical resistivity logs are the oil industry's basic tool for the in-situ determination of hydrocarbon saturation. For clean sands, Archie's empirical equations (1942) have provided the basis for the fluid saturation calculations. In shaly sands, however, exchange counterions associated with clay minerals increase rock conductivity over that of clean sand, and the Archie relations are no longer valid. For shaly sands containing dispersed clay, the Waxman-Smits equation has proven successful in accounting for the additional clay conductance with measurable petrophysical parameters (Waxman and Smits, 1968; Waxman and Thomas, 1974).

In order to use the Waxman-Smits equations, the shaliness parameter Q_v , defined as the cation exchange capacity of the sandstone per unit pore volume, must be obtained. However, Q_v cannot be determined directly from present wireline logs, although various indirect techniques have been proposed using core-derived empirical relations in conjunction with log responses (Lavers et al., 1975; Juhasz, 1979, 1981). If core samples or drill cuttings are available, Q_v can be measured in the laboratory by (1) a *destructive method* using standard cation exchange capacity, grain density, and porosity determinations (Waxman and Smits, 1968) and (2) a *nondestructive method* using membrane potential measurements (Thomas, 1976).

The Waxman-Smits equations apply only to in-phase conductivity, that is, conduction which occurs in phase with the reference current. Electrical impedance is, however, a complex quantity with a reactive as well as a resistive component. Out-of-phase or quadrature conduction at low frequencies is commonly referred to as induced polarization (IP).

Recently, attention has been focused on induced polarization in shaly sands as a means of determining cation exchange capacity (Snyder et al., 1977; Sumner, 1976; Sill, 1964; Hoyer and Rumble, 1976). Although these workers have observed that IP is proportional to shaliness, what has been lacking is a model which relates IP response in shaly sands to basic petrophysical quantities, such as electrolyte concentration and oil saturation.

We present equations for the out-of-phase conductivity of shaly sands, consistent with the Waxman-Smits equations and using the same petrophysical variables. Unlike in-phase conductivity, the out-of-phase response is a weak function of salinity, predictable from clay counterion and electrolyte ion mobilities. The IP equations can be solved simultaneously with the Waxman-Smits equations for one of the variables which is often unknown, for example, shaliness or electrolyte conductance. Then hydrocarbon saturation can be determined independent of shaliness or the salinity of the formation waters.

The organization of this paper is as follows. The first section reviews parameters used to describe induced polarization. The model for out-of-phase (quadrature) conductivity is presented in the second section. Predictions from the model are compared with laboratory measurements of shaly sand quadrature conductivities as functions of shaliness and sodium chloride electrolyte concentration (the third section), and oil saturation (the final section). Apparatus and experimental procedures are described in Appendices A and B.

INDUCED-POLARIZATION PARAMETERS

For a linear, homogeneous, isotropic and time-invariant medium, the general constitutive equations are

$$\mathbf{J} = C^*(\omega)\mathbf{E} = [C_I(\omega) + iC_Q(\omega)]\mathbf{E},$$

and

$$\mathbf{D} = K^*(\omega)\epsilon_0 \mathbf{E} = [K_I(\omega) + iK_Q(\omega)]\epsilon_0 \mathbf{E},$$

where $\mathbf{J}$ is the current density, $\mathbf{D}$ the displacement field, $\mathbf{E}$ the electric field, C^* the complex conductivity, K^* the complex dielectric constant, and ϵ_0 the permittivity of free space (8.85×10^{-12} F/m). $\{C_I, C_Q\}$ and $\{K_I, K_Q\}$ are the in-phase and quadrature components of C^* and K^* , respectively. For linear, causal systems, $\{C_I, C_Q\}$ and $\{K_I, K_Q\}$ each satisfy the Kramers-Kronig relations, and are in general frequency-dependent.

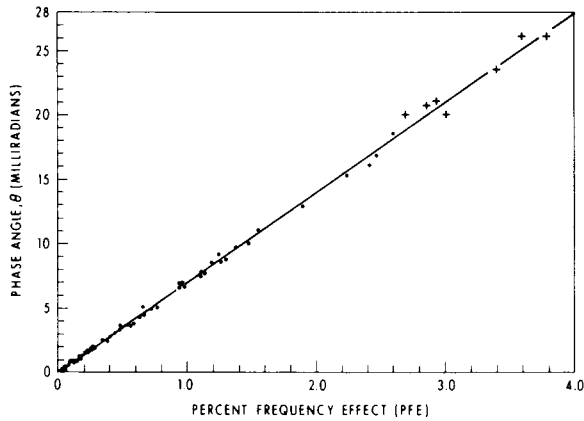


FIG. 1. Relationship between phase angle and percent frequency effect (PFE).

The phase angle θ is defined as the angle between in-phase and quadrature conductivities:

$$\theta(\omega) = \tan^{-1} \frac{C_Q(\omega)}{C_I(\omega)} \cong \frac{C_Q}{C_I} \quad \text{for small } \theta. \quad (2)$$

When the constitutive relations (1) are incorporated into the Maxwell equation for the curl of the magnetic field $\mathbf{H}$, i.e.,

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \quad (3)$$

and assuming harmonic excitation of the electric field, $\mathbf{E} = \mathbf{E}_0 e^{i\omega t}$, then

$$\nabla \times \mathbf{H} = [C_{\text{eff}} + i\omega\epsilon_0 K_{\text{eff}}]\mathbf{E}, \quad (4)$$

where C_{eff} and K_{eff} are the "effective" conductivity and dielectric constant:

$$C_{\text{eff}}(\omega) = C_I(\omega) - \omega\epsilon_0 K_Q(\omega), \quad (5)$$

$$K_{\text{eff}}(\omega) = \frac{C_Q(\omega)}{\omega\epsilon_0} + K_I(\omega).$$

As indicated by Fuller and Ward (1970), $C_{\text{eff}}(\omega)$ and $K_{\text{eff}}(\omega)$ are the electrical quantities actually measured on rocks.

In the following discussion, the general equations (5) are specialized to the case of brine-saturated rocks with connected pore spaces, measured at low frequencies. Under these conditions, ionic conduction is expected to be the dominant conduction mechanism and dielectric loss is therefore neglected. Following Fuller and Ward, we assume

$$C_I \gg \omega\epsilon_0 K_Q \quad \text{and} \quad C_{\text{eff}} \cong C_I. \quad (6)$$

Also, if C_Q is nearly independent of frequency, then at low frequencies in shaly sands

$$\frac{C_Q}{\omega\epsilon_0} \gg K_I \quad \text{and} \quad K_{\text{eff}} \cong \frac{C_Q}{\omega\epsilon_0}. \quad (7)$$

This is the basis for observations by Hoyer and Rumble (1976)

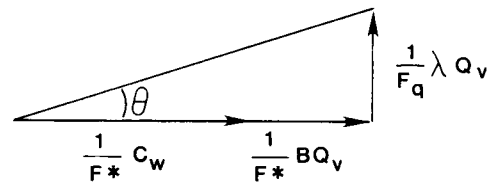
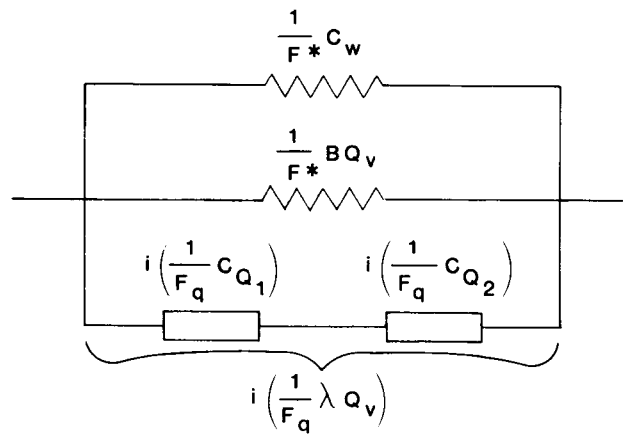


FIG. 2. Complex conductance model for shaly sands.

that K_{eff} increases as $1/\omega$ in shaly sands for frequencies below 10^5 Hz.

The complex resistivity ρ^* is defined as

$$\rho^*(\omega) = \rho_I(\omega) + i\rho_Q(\omega) = 1/C^*(\omega).$$

Frequency effect (FE) is the fractional rate of change of $\rho_I(\omega)$ between two low frequencies, ω and $A\omega$

$$\text{FE} \equiv \frac{\rho_I(\omega) - \rho_I(A\omega)}{\rho_I(\omega)}. \quad (8)$$

For phase angles which are slowly varying functions of frequency, FE is proportional to phase shift. In particular, Shuey and Johnson (1973) and Van Voorhis et al. (1973) have shown that the relationship between FE and θ at the geometric mean frequency $\sqrt{A\omega}$ is

$$\text{FE} = \left(1 - A \frac{2}{\pi} \theta\right) \cong -\frac{2}{\pi} \theta \log_e A, \quad \theta \ll 1. \quad (9)$$

When a frequency difference of one decade is chosen, $A = 10$; the percent frequency effect (PFE) then is

$$\text{PFE} \equiv 100 \times \text{FE} = 0.1466\theta \quad (\text{mrad}). \quad (10)$$

Equation (10) may be used to check complex resistivity measurements for self-consistency. In Figure 1 phase shift and PFE are plotted for our twenty sandstone samples at five different salinities, as reported in the third section. The PFE was computed between 10 and 100 Hz, and the phase angle θ was measured near the geometric mean frequency (30 Hz). The crosses in Figure 1 are field data from Van Voorhis et al. (1973); their PFE was taken between 0.1 and 1.0 Hz. A linear regres-

sion of PFE on θ using our sandstone data gives

$$\text{PFE} = [0.144 \pm 0.003]\theta \quad (\text{mrad}),$$

in agreement with equation (10).

Because FE is typically a few percent or less in shaly sands, the frequency variation of $C_I(\omega)$ will not be shown explicitly in the following discussion. Nevertheless, $C_I(\omega)$ satisfies the Kramers-Kronig relations and has the frequency dependence shown in equation (10).

THE MODEL

The complex conductance model can be represented by the equivalent circuit shown in Figure 2. We describe first the in-phase and then the quadrature elements.

In-phase conductivity model

The Waxman-Smits model (1968) consists of two conductive elements in parallel, one consisting of the conductance of free electrolyte contained in the pores and the other the conductance contributed by the exchange cations of the clay surfaces (Figure 2). Waxman and Smits proposed the following equation for the in-phase conductivity of brine-saturated shaly sands:

$$C_I = \frac{1}{F^*} (C_w + BQ_v), \quad (11)$$

where

C_I = specific in-phase conductivity of water-bearing shaly sand (mho-cm^{-1}),¹

F^* = formation resistivity factor defined by Waxman-Smits,

C_w = specific conductivity of the equilibrating electrolyte solution (mho-cm^{-1}),

Q_v = cation exchange capacity of the rock/unit pore volume (meq-ml^{-1}),

and

B = equivalent ionic conductance of clay exchange cations as function of C_w ($\text{mho-cm}^2\text{-meq}^{-1}$).

From their data, Waxman and Smits obtained an empirical expression for B :

$$B = 0.0383[1 - 0.83 \exp(-50C_w)] \quad (\text{mho-cm}^2\text{-meq}^{-1}). \quad (12)$$

The constants in equation (12) are for shaly sand in equilibrium with NaCl solution at 25°C. Equation (12) shows that the equivalent ionic conductance of the Na^+ exchange cation increases with increasing electrolyte conductivity up to a maximum value of $B_{\max} = 0.0383 \text{ mho-cm}^2\text{-meq}^{-1}$. $B_{\max}$, as well as the other parameters in equation (12), depends upon cation type.

For oil-bearing shaly sands, Waxman and Smits proposed the equation

$$C_I = \frac{S_w^{n^*}}{F^*} (C_w + BQ_v/S_w), \quad (13)$$

¹In this paper, C_I and C'_I denote in-phase conductivities for water- and oil-bearing sands. Notations were C_0 and C_i respectively in previous papers (Waxman and Smits, 1968; Waxman and Thomas, 1974).

where

C'_I = specific in-phase conductance of oil-bearing sand (mho-cm^{-1}),

S_w = fractional water saturation,

and

n^* = saturation exponent.

Equation (13) is developed from equation (11) with the additional assumptions that the formation factor for oil-bearing sands is $F^*/S_w^{n^*}$ and the volume concentration of clay exchange cations is $Q'_v = Q_v/S_w$. The validity of equation (13) was demonstrated by Waxman and Thomas (1974).

Quadrature conductivity model

The mechanism proposed here for the quadrature conductivity is similar to that of Madden and Marshall (1959), where the rock pore is modeled as an alternating series of clay-rich and clay-free zones. The clay-rich zones have high concentrations of fixed negative charge sites which enhance cation relative to anion transport. In the steady-state condition after constant current is applied, the inequality of cation and anion flows causes local concentration excesses and deficiencies to form at clay sites, alternating throughout the pore length. When the applied current is terminated, the diffusion voltages resulting from these concentration gradients decay with time as the ions redistribute themselves to their equilibrium positions.

The general solution for the equations of motion of each ion species subject to diffusion and electric forces shows two contributions to the local concentration changes (Madden and Marshall, 1959; Buchheim and Irmer, 1979). The first contribution is a space charge developed at the boundaries of a clay-rich zone, which, at low frequencies, has a characteristic length equal to the Debye-Hückel distance. The second contribution is from ionic diffusion to the clay-rich zones, which act as cation-selective membranes for electrolyte transport.

In this spirit, Figure 2 models the induced concentration gradients as arising from two mechanisms: (1) clay counterion displacement, and (2) electrolyte blockage by the clay-site membranes. Both mechanisms are assumed to be proportional to Q_v , i.e.,

$$\Delta c_1 = A_1 Q_v \quad \text{and} \quad \Delta c_2 = A_2 Q_v, \quad (14)$$

where $\Delta c_{1,2}$ are the concentration excesses, and $A_{1,2}$ are the proportionality constants for each mechanism. We first consider the specific quadrature conductivity in a straight section of pore; the tortuosity and geometry factors are added later through the quadrature formation factor. Multiplying Δc_1 by the equivalent conductance of the clay counterions B gives C_{Q1} , the specific quadrature conductance of the displaced clay exchange cations as

$$C_{Q1} = A_1 B Q_v. \quad (15)$$

Similarly, multiplying Δc_2 by the equivalent conductance of electrolyte solution Λ gives C_{Q2} , the specific quadrature conductance of the blocked electrolyte, as

$$C_{Q2} = A_2 \Lambda Q_v. \quad (16)$$

In order to account for membrane efficiency on C_{Q2} , equation (16) is multiplied by the factor $(2t_+ - 1)$, where t_+ is the cation transport number through the clay-rich section, i.e.,

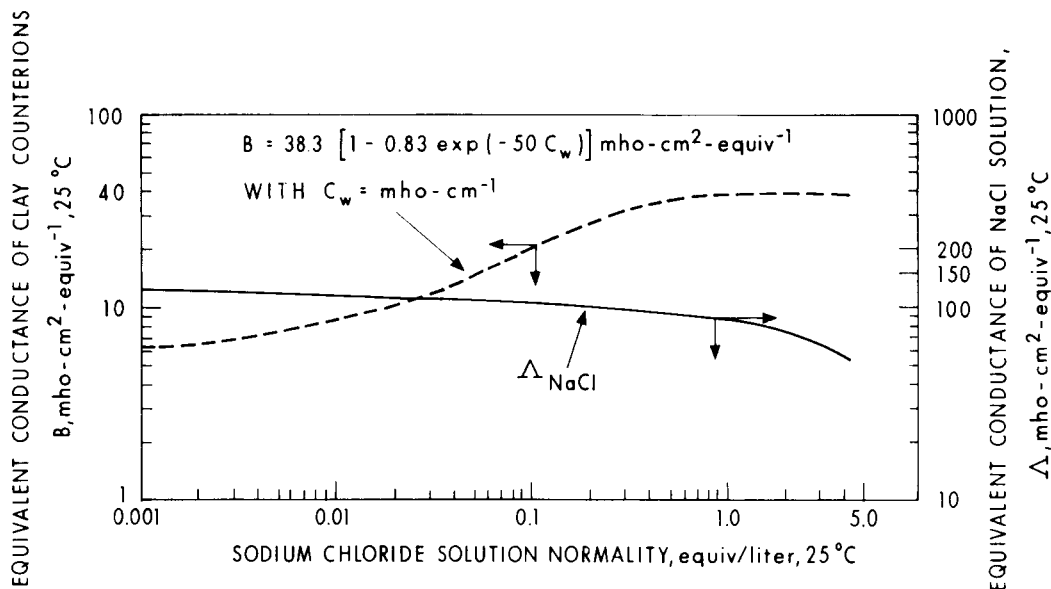


FIG. 3. Equivalent conductances of NaCl solution and sodium exchange cations versus NaCl concentration at 25°C.

$$C_{Q_2} = A_2(2t_+ - 1)\Lambda Q_v. \quad (17)$$

If no transport of anions occurs through the clay-rich zones (perfect membrane behavior), $t_+ = 1$ and equation (17) reduces to equation (16).

In the complex quadrature model in Figure 2, the quadrature conductances C_{Q_1} and C_{Q_2} are modeled in series. Thus

$$\frac{1}{(C_Q)_{\text{series}}} = \frac{1}{C_{Q_1}} + \frac{1}{C_{Q_2}}, \quad (18)$$

or

$$(C_Q)_{\text{series}} = \frac{A_1 A_2 (2t_+ - 1) \Lambda B}{[A_1 B + A_2 (2t_+ - 1) \Lambda]} Q_v. \quad (19)$$

An equivalent quadrature conductance λ is defined by

$$\lambda = \frac{(A_1 B)[A_2 (2t_+ - 1) \Lambda]}{[A_1 B + A_2 (2t_+ - 1) \Lambda]}, \quad (20)$$

or for t_+ equal to unity,

$$\lambda = \frac{A_1 B A_2 \Lambda}{A_1 B + A_2 \Lambda}, \quad (21)$$

so that

$$(C_Q)_{\text{series}} = \lambda Q_v. \quad (22)$$

The behavior of Λ_{NaCl} and B (sodium exchange cation) as a function of sodium chloride solution concentration is shown in Figure 3. Because Λ_{NaCl} and B are weak functions of salt concentration, λ will also have a weak dependence on salinity.

The effects of pore cross-sectional area and tortuosity (effective path length) are now considered. For in-phase conductivity, a reduction in pore cross-sectional area or an increase in tortuosity reduces the mean current flow through the rock. Both effects are usually included in the formation resistivity

factor, which is F for the Archie clean-sand case and F^* in the Waxman-Smiths notation.

Empirical studies by Archie (1942) indicated that the formation resistivity factor may be represented by $F = \phi^{-m}$ where ϕ is the fractional porosity and the cementation factor m is approximately equal to 2 for consolidated rocks. For shaly sands, the analogous Waxman-Smiths notations are F^* and m^* . Since the quadrature current travels along the same tortuous path as the current attributed to the ions in the pore water, C_Q should depend upon the same formation resistivity factor F^* as does the Waxman-Smiths equation. That is, if F_q is the formation factor for quadrature conductivity, then we have

$$F_q = F^*. \quad (23a)$$

However, quadrature conductivity may have additional dependence upon pore size because constriction of pores increases both the membrane blockage effect and displacement of clay counterions. In the next section we show that, although equation (23a) represents a good empirical fit to our data for typical reservoir rocks, a better fit is obtained with

$$F_q = F^* \phi. \quad (23b)$$

This form for the quadrature formation factor is used throughout this paper.

The general equation for the quadrature conductivity of water-saturated shaly sands then becomes

$$C_Q = \frac{1}{F_q} \lambda Q_v. \quad (24)$$

When the sand is oil-bearing, the formation factor F'_q (primes indicate presence of oil) will increase according to Archie's second empirical equation (1942) and we obtain

$$F'_q = F_q S_w^{-n^*}. \quad (25)$$

Waxman and Thomas (1974) showed that in oil-bearing sands

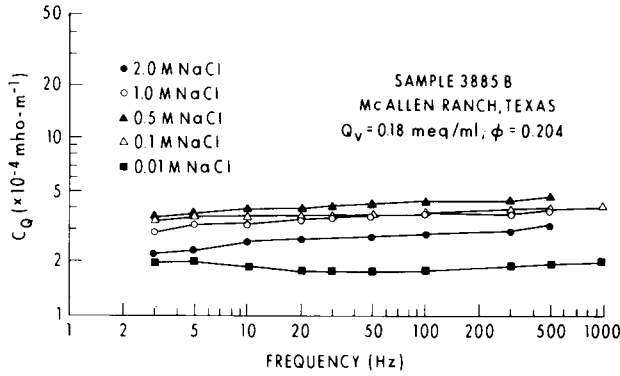


FIG. 4. Frequency independence of quadrature conductivity, sample 3885B.

the shaliness parameter Q'_v is increased by

$$Q'_v = Q_v / S_w \quad (26)$$

Substitution of equations (25) and (26) into equation (24) yields the equation for quadrature conductivity of oil-bearing shaly sands as

$$C'_Q = \frac{1}{F_q} \lambda Q_v S_w^{n^*-1} \quad (27)$$

When equations (2), (11), and (24) are combined, the phase angle in brine-saturated sands is

$$\theta = \frac{C_Q}{C_I} = \frac{\lambda Q_v}{\phi(C_w + BQ_v)} \quad (28)$$

Figure 2 shows the vectorial relations between C_I and C_Q . Similarly, when equations (2), (13), and (27) are combined, the

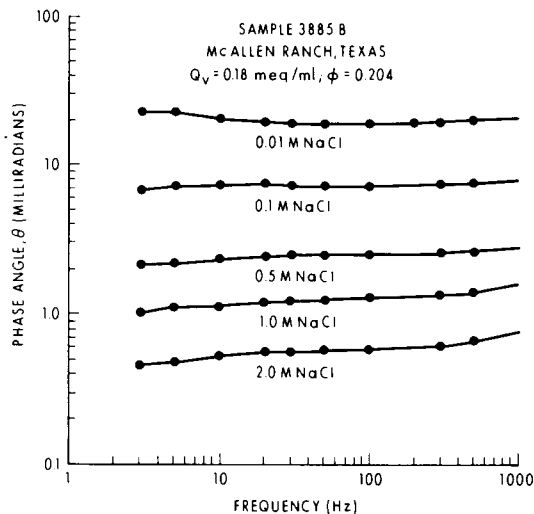


FIG. 5. Frequency independence of phase angle, sample 3885B.

phase angle θ' in oil-bearing shaly sands is

$$\theta' = \frac{C'_Q}{C'_I} = \frac{\lambda Q_v}{\phi S_w \left[C_w + \frac{BQ_v}{S_w} \right]} \quad (29)$$

The ratio of phase angles is then

$$\frac{\theta}{\theta'} = \frac{S_w [C_w + (BQ_v/S_w)]}{(C_w + BQ_v)} \quad (30)$$

Solving equation (30) for S_w leads to

$$S_w = \frac{\theta}{\theta'} \left[1 + \frac{BQ_v}{C_w} \right] - \frac{BQ_v}{C_w} \quad (31)$$

At high salinities and/or in clean sands, i.e., $BQ_v \ll C_w$, equation (31) reduces to

$$S_w = \frac{\theta}{\theta'} \quad (32)$$

Thus, measurements of the phase angles θ and θ' permit the determination of oil/water saturations without knowledge of the Waxman-Smiths saturation exponent n^* , if it is assumed that only S_w varies in traversing from brine-saturated to oil-bearing zones in the same shaly sand formation.

In summary, the model predicts that quadrature conductivity is (1) directly proportional to the shaliness parameter Q_v , (2) weakly dependent on salinity, and (3) reduced in oil-bearing sands by the factor $S_w^{n^*-1}$.

Frequency dependence and limits of applicability

The distance an ion is displaced by the electric field determines the time required for the ion to return to its equilibrium position. Therefore, the spectrum of relaxation distances determines the frequency dependence of the quadrature conductance. The broad distribution of relaxation lengths in naturally occurring sands results in quadrature conductance essentially independent of frequency over a wide range of frequencies.

For all our samples, the quadrature conductivities and phase angles were approximately independent of frequency in the range 3–1000 Hz, as shown in Figures 4 and 5 for a typical sand. Since C_Q is frequency-independent in this range, the effective dielectric constant K_{eff} increases as $1/\omega$ when frequency is lowered [equation (7)]. This was also reported by Hoyer and Rumble (1976), as well as Poley et al. (1978).

Frequency-independent IP behavior cannot be valid over all frequencies, however. The frequency limits f_{min} and f_{max} of the frequency-independent IP spectrum can be estimated from the diffusion constant D of sodium and chloride ions in aqueous solution (Jost, 1952) as

$$D \sim \frac{r^2}{t} = 1.5 \times 10^{-5} \text{ cm}^2/\text{s}, \quad (33)$$

where r is displacement length (cm) and t is relaxation time (s).

For frequencies below f_{min} , the ions have ample time to diffuse across the longest clay-free zones and larger concentration gradients do not develop. A scale for longest relaxation length is set by the distance between adjacent but not connected clay aggregates. If the clay occurs as a uniform grain coating, then this distance should be the largest intergrain

distance, approximately 100 microns. This yields $t \sim 6.7$ s and $f_{\min} = 1/(2\pi t) \sim 0.024$ Hz. For higher clay concentrations or smaller pore sizes, $f_{\min}$ may be higher than this estimate. As discussed later, this partly explains IP behavior in shales.

A general mathematical argument shows that $C_Q(\omega)$ must approach zero at low enough frequency (Fuller and Ward, 1970). Conductance is a physical response function, so the imaginary part of its Fourier transform, $C_Q(\omega)$, is an odd function of ω . $C_Q(\omega)$ must therefore go to zero at $\omega = 0$. Moreover, if the effective dielectric constant in equation (5) is to remain

finite as $\omega \rightarrow 0$, $C_Q(\omega)$ must approach zero at least as fast as $\omega^{1.0}$ at the low-frequency end, as expected for capacity-limited behavior.

Diffusion-limiting occurs at the high-frequency end of the IP spectrum. At frequencies above $f_{\max}$, the ions cannot diffuse fast enough to develop concentration gradients over the shortest relaxation distances. If the microporous regions between clay aggregates are ~ 100 Å, then $f_{\max} \sim 2.4 \times 10^6$ Hz. The shortest possible spatial variation between charged and uncharged zones is the average distance between charge sites on a clay

Table 1. Petrophysical properties of the samples.

Sample	Number	Porosity ϕ (Percent)	Brine Permeability* k (md)	Q_v meq/ml	X-Ray Weight Percent Clay	Clay Type
1. Benton Field, IL	3477	20.1	833	.034	2	Kaolinite with Chlorite
2. North Slope, AK	3336A	20.5	71.7	.042	10	Kaolinite
3. Benton Field, IL	3478	18.1	78.2	.055	6	Kaolinite with Chlorite
4. Berea Outcrop	101	21.0	?	.055	10	Kaolinite with Illite
5. Berea Outcrop	102	20.5	?	.064	10	Kaolinite with Illite
6. Crown Zell, Tuscaloosa	CZ10	23.3	116	.084	~ 5	Chlorite
7. Little Creek Pike County, MI	3833A	26.8	118	.10	22.4	Chlorite
8. Bay Marchand, Blk 2 Offshore, LA	3126B	24.9	468	.12	~ 5	Illite, Montmorillonite, Chlorite, Kaolinite
9. Camp Shelby Outpost, MI	3847A	12.6	.66	.14	~ 10	Illite, Chlorite with Kaolinite
10. Helen Gohlke, Dewitt County, TX	3283A	18.6	40	.16	?	Kaolinite, Chlorite, Illite
11. McAllen Ranch, TX	3885B	20.4	4.95	.18	~ 25	Illite with Chlorite
12. Gulf of Alaska	3972E	18.7	407	.185	5	Chlorite, Montmorillonite with Illite
13. Provident City, TX	3258A	12.8	0.61	.30	~ 18	Illite
14. McAllen Ranch, TX	3891A	19.2	0.26	.385	~ 32	Chlorite with Illite
15. Big Foot, Frio County, TX	3308A	27.0	5.0	.409	~ 35	Montmorillonite, Chlorite with Kaolinite, Illite
16. Big Foot, Frio County, TX	3323F	28.8	3.1	.59	?	Same as 15
17. Big Foot, Frio County, TX	3324A	22.0	1.48	.65	?	Same as 15
18. Big Foot, Frio County, TX	3323E	28.1	1.26	.67	?	Same as 15
19. Big Foot, Frio County, TX	3324B	20.4	1.14	.84	?	Same as 15
20. Big Foot, Frio County, TX	3306F	17.1	.70	.95	?	Same as 15
21. Big Foot, Frio County, TX	3323C	29.6	3.0	.55 [#] .61 [§]	35	Montmorillonite, Chlorite with Kaolinite, Illite

*100 grams NaCl per liter.

[#]Analytical determination.

[§]Multiple Salinity Method - Conductivity Plot.

platelet, about 10 Å. For montmorillonite this distance is 8.25 Å and it does not vary by more than a factor of 1.5 between clay mineral species (Van Olphen, 1977). Then $t \sim 6.7 \times 10^{-10}$ s and $f_{\max} \sim 2.4 \times 10^8$ Hz. Clay effects will therefore be a significant component of the effective dielectric constant even at radio frequencies.

Equation (24) indicates that C_Q should continue to increase as the Q_v parameter of the rock increases. Experiments have shown, however, that clay-rich shales do not exhibit induced polarization at low frequencies (Madden and Marshall, 1959; Dakhnov et al., 1952; Dakhnov, 1959). Therefore, the IP model becomes invalid somewhere in the transition from shaly sands to shales. Where this transition zone begins is not known, but it is clear from data presented in the next section that the model applies for typical reservoir sands for $Q_v \leq 1.0$ meq/ml and $\phi \geq 0.12$.

The model does not apply in the case of shales for at least two reasons.

(1) The abundance of clay fixed-charge sites amounts to a clay conduction path which is spatially continuous. Since there are few clay-free pores, concentration gradients do not develop in the presence of the applied electric field.

(2) The relaxation time, which is determined by the length of the clay-free zones, is so short that measurable polarization should exist only at higher frequencies. We have observed a steady increase of polarization in samples of Pierre shale as the frequency was increased to 10 kHz.

IP MEASUREMENTS OF SHALINESS AND ELECTROLYTE CONCENTRATION

This section describes laboratory measurements of IP in brine-saturated shaly sands.

Equations (20) and (21) define the equivalent quadrature conductance λ in terms of the equivalent conductance of sodium chloride solution Λ , the equivalent conductance of clay exchange cations B , and A_1 and A_2 , which are parameters for each sandstone. The term $A_1 B$ arose from displacement of the counterions associated with the clay minerals by the applied current and the term $A_2 \Lambda$ from electrolyte blockage in the clay-rich zones. A_1 and A_2 are determined by measuring quadrature conductivity in typical shaly sands. In particular, we have obtained an average value of λ which applies to all sands (denoted as $\bar{\lambda}$ with parameters $\bar{A}_1$ and $\bar{A}_2$) and which is a good approximation in equations (20) and (21).

Experimental procedures

Complex conductivity measurements were made on twenty sandstone plugs (obtained from eleven formations) at five different salinities: 2.0, 1.0, 0.5, 0.1, and 0.01 M NaCl. Petrophysical and geologic descriptions of the sandstones are in Table 1. The samples varied widely in cation exchange capacity and Q_v ($0.03 \leq Q_v \leq 0.95$ meq/ml), type of clay, and clay distribution in

Table 2.1. In-phase and quadrature conductivities (30 Hz, 25°C) with supporting petrophysical data for shaly sand samples.

Sample	Salinity	C_1 (mho-m ⁻¹)	C_Q (10 ⁻⁴ mho-m ⁻¹)	θ (Milli-radians)	Q_v Waxman-Smiths meq/ml	Q_v Analysis	F*	ϕ (%) Porosity	$C_Q F^* \phi$ (10 ⁻⁴ mho-m ⁻¹)
3477	2 M	.9624	2.06	.21	.034	.026*	15.43	20.1	6.39
	1 M	.5630	1.50	.27					4.65
	0.5 M	.3144	1.10	.35					3.41
	0.1 M	.0741	.669	.90					2.07
	0.01 M	.0109	.418	3.84					1.30
3336A	2 M	.6281	1.73	.28	.042	.042*	24.10	20.5	8.55
	1 M	.3579	1.17	.33					5.78
	0.5 M	.1985	.930	.47					4.59
	0.1 M	.0526	.806	1.53					3.98
	0.01 M	.00741	.651	8.78					3.22
3478	2 M	.7472	1.71	.23	.055	.052*	20.08	18.1	6.21
	1 M	.4407	1.41	.32					5.12
	0.5 M	.2382	1.26	.53					4.58
	0.1 M	.0640	1.09	1.70					3.96
	0.01 M	.00832	.752	9.03					2.73
101	2 M	.8602	2.14	.25	.068	.055	17.51	21.0	7.87
	1 M	.5119	-	-					-
	0.5 M	.2826	-	-					-
	0.25 M	.1580	-	-					-
	0.1 M	.0713	1.88	2.64					6.91
	0.05 M	.0403	1.76	4.36					6.47
	0.01 M	.0113	1.14	10.1					4.19
102	1 M	.4909	1.29	.26	.070	.064	17.95	20.5	4.75
	0.5 M	.2789	1.57	.56					5.78
	0.25 M	.1530	1.82	1.19					6.70
	0.1 M	.0695	1.85	2.66					6.81
	0.05 M	.0384	1.69	4.39					6.22
	0.01 M	.00845	.833	9.86					3.07

*From adjacent plugs.

the rock. The samples contained only trace amounts of conductive metallic minerals.

Plugs were 1 inch in diameter and between $1\frac{1}{2}$ and 2 inches in length. Each sample was flushed with a 3:1 azeotrope of chloroform/methanol to remove all hydrocarbons, then with methanol to remove all soluble salts.

Standard core analyses of these samples yielded values for bulk volume, grain volume, porosity, grain density, and permeability to air. Values of Q_v were determined by the destructive analytical technique described by Waxman and Smits (1968). Cation exchange capacities were determined by the procedure described by Mortland and Mellor (1954).

Since shaly sands often disintegrate in low-salinity brines, the conductance measurements were started in 2.0 M NaCl and salinity was reduced in sequence. The plugs were saturated under vacuum with 2.0 M NaCl solution for 24 hours. Very shaly sandstones were equilibrated with this solution for several months before conductivities were recorded. Electrolyte concentration was changed by slowly flushing each plug with 2 liters (about 400 pore volumes) of the next, less-concentrated NaCl solution in the series. A conductivity cell was used to check to 1 percent accuracy the salinities of the input and effluent brines. Conductivities for the very shaly Big Foot sandstones were not obtained at 0.01 M NaCl due to sample disaggregation.

After measurements were completed at 0.01 M NaCl, several samples were reequilibrated at higher salinities; typically values of C_I and C_Q were repeatable to 2 percent or better.

In Appendices A and B, experimental design considerations, sample IP cells, electrodes, the frequency-domain bridge circuit, and tests of IP measurement accuracy are described. The accuracy of the measurements is ± 1.0 percent after correction for temperature and cell constants. Measurements were made from 3 to 1000 Hz.

Measurements were carried out at ambient laboratory temperatures, typically $25 \pm 1^\circ\text{C}$. Small temperature corrections to conductivities were made to obtain C_I and C_Q at 25°C , using Arps' formula (1953). Some conductivity measurements were made at temperatures up to 100°C with the entire IP sample cell immersed in a temperature-controlled oil bath.

Presentation of data

In-phase and quadrature conductivities at 25°C and 30 Hz for the twenty brine-saturated sandstones are given in Table 2. Plots of C_I versus the equilibrating brine conductivity C_w were made for each plug; a typical example is in Figure 6. Values of F^* and $(Q_v)_{WS}$ in Table 2 were computed from the slopes and intercepts, respectively, of the straight-line portions of the C_I versus C_w plots using the Waxman-Smits (WS) multiple-salinity method (1968). A value of $B_{\max}$ equal to $38.3 \text{ mho-cm}^2\text{-meq}^{-1}$ was used in computing $(Q_v)_{WS}$. Also included in Table 2 are the Q_v values determined by the analytical procedures described previously, obtained either by destructive measurement on the particular sample or, where indicated, by measurement on adjacent plugs.

Table 2.2. In-phase and quadrature conductivities (30 Hz, 25°C) with supporting petrophysical data for shaly sand samples.

Sample	Salinity	C_I (mho-m ⁻¹)	C_Q (10 ⁻⁴ mho-m ⁻¹)	θ (Milli- radians)	Q_v Waxman-Smits meq/ml	Q_v Analysis	F^*	$\phi(\%)$ Porosity	$C_Q F^* \phi$ (10 ⁻⁴ mho-m ⁻¹)
C210	2 M	.8110	2.79	.34	.084	.07*	18.83	23.3	12.24
	1 M	.4723	2.13	.45					9.35
	0.5 M	.2689	1.73	.64					7.59
	0.1 M	.0698	1.33	1.90					5.84
	0.01 M	.0138	.935	6.78					4.10
3833A	2 M	1.0106	3.57	.35	.10	.110*	15.38	22.4	12.30
	1 M	.6240	2.23	.36					7.68
	0.5 M	.3246	1.95	.60					6.72
	0.1 M	.0932	1.54	1.65					5.31
	0.01 M	.0144	.997	6.92					3.43
3I26B	2 M	1.1999	5.13	.43	.14	.120	12.82	24.9	16.38
	1 M	.7193	4.76	.66					15.19
	0.5 M	.4094	4.36	1.07					13.92
	0.1 M	.1045	3.89	3.72					12.42
	0.01 M	.0195	2.14	11.0					6.83
3847A	1 M	.1704	1.42	.83	.136	.144	53.48	12.6	9.57
	0.5 M	.0944	1.73	1.81					11.66
	0.25 M	.0545	1.80	3.32					12.13
	0.1 M	.0269	1.77	6.58					11.93
	0.05 M	.0159	1.60	10.05					10.78
3283A	0.01 M	.00525	1.06	20.2					7.14
	2 M	.6906	3.20	.46	.159	.15*	22.57	18.6	13.43
	1 M	.4466	3.43	.77					14.40
	0.5 M	.2376	3.74	1.57					15.70
	0.1 M	.0675	3.45	5.11					14.48
	0.01 M	.0103	1.74	16.9					7.30

*From adjacent plugs.

Table 2.3. In-phase and quadrature conductivities (30 Hz, 25°C) with supporting petrophysical data for shaly sand samples.

Sample	Salinity	C_I (mho-m ⁻¹)	C_Q (10 ⁻⁴ mho-m ⁻¹)	θ (Milli-radians)	$\frac{Q_V}{\text{Waxman-Smits}}$ meq/ml	$\frac{Q_V}{\text{Analysis}}$	F*	ϕ (%) Porosity	$C_Q F^* \phi$ (10 ⁻⁴ mho-m ⁻¹)
3885B	2 M	.4767	2.76	.58	.157	.180	32.36	20.4	18.22
	1 M	.2838	3.59	1.27					23.70
	0.5 M	.1675	4.16	2.48					27.46
	0.1 M	.0492	3.68	7.48					24.29
	0.01 M	.00889	1.76	19.8					11.62
3972E	2 M	.6597	3.56	.54	.11	.185	23.31	18.7	15.52
	1 M	.3866	3.52	.91					15.34
	0.5 M	.2169	3.38	1.56					14.73
	0.1 M	.0564	2.47	4.38					10.77
	0.01 M	.00983	1.28	13.0					5.58
3258A	2 M	.2790	2.47	.89	.30	.372*	58.82	12.8	18.60
	1 M	.1685	3.69	2.19					27.78
	0.5 M	.1016	4.17	4.10					31.40
	0.1 M	.0364	3.50	9.61					26.35
	0.01 M	.0103	1.58	15.3					11.90
3891A	2 M	.3668	2.74	.75	.40	.385	43.67	19.2	22.97
	1 M	.2354	4.36	1.85					36.56
	0.5 M	.1442	5.33	3.70					44.69
	0.1 M	.0547	4.64	8.48					38.90
	0.01 M	.0240	2.19	9.12					18.36
3308A	2 M	1.0768	8.15	.76	.384	.409	15.24	27.0	33.54
	1 M	.6587	10.8	1.64					44.44
	0.5 M	.4002	12.4	3.10					51.02
	0.1 M	.1446	11.2	7.75					46.09
	0.01 M	.0443	4.90	11.1					20.16
3323F	2 M	1.0969	7.52	.69	.59	.69*	15.67	28.8	33.94
	1 M	.6920	12.0	1.74					54.16
	0.5 M	.4404	14.6	3.32					65.89
	0.1 M	.1789	12.4	6.91					55.96

*From adjacent plugs.

Table 2.4. In-phase and quadrature conductivities (30 Hz, 25°C) with supporting petrophysical data for shaly sand samples.

Sample	Salinity	C_I (mho-m ⁻¹)	C_Q (10 ⁻⁴ mho-m ⁻¹)	θ (Milli-radians)	$\frac{Q_V}{\text{Waxman-Smits}}$ meq/ml	$\frac{Q_V}{\text{Analysis}}$	F*	ϕ (%) Porosity	$C_Q F^* \phi$ (10 ⁻⁴ mho-m ⁻¹)
3324A	2 M	.5647	6.25	1.11	.80	.646	32.26	22.0	44.36
	1 M	.3808	9.53	2.50					67.64
	0.5 M	.2547	11.5	4.53					81.62
	0.1 M	.1268	9.87	7.79					70.05
3323E	2 M	1.0534	9.69	.92	.67	.69*	16.56	28.1	45.09
	1 M	.6728	13.5	2.00					62.82
	0.5 M	.4376	15.5	3.53					72.13
	0.1 M	.1807	12.1	6.69					56.31
3324B	2 M	.5699	7.34	1.29	.94	.842	32.47	20.4	48.62
	1 M	.3776	10.6	2.81					70.21
	0.5 M	.2560	12.8	5.00					84.79
	0.1 M	.1247	10.7	8.60					70.88
3306F	2 M	.3898	6.86	1.76	.95	.95	47.17	17.1	55.33
	1 M	.2681	10.9	4.07					87.92
	0.5 M	.1781	13.0	7.30					104.86

*From adjacent plugs.

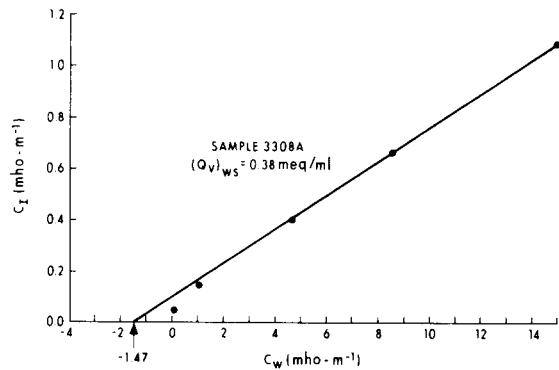


FIG. 6. In-phase sand conductivity versus NaCl conductivity, sample 3308A.

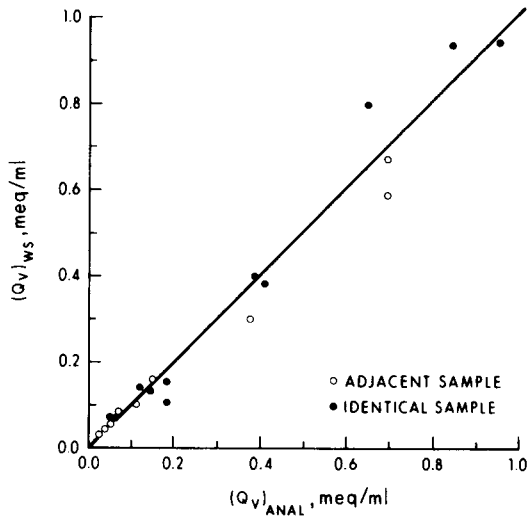


FIG. 7. Regression of Q_v from Waxman-Smiths equation on Q_v from exchange capacity measurements.

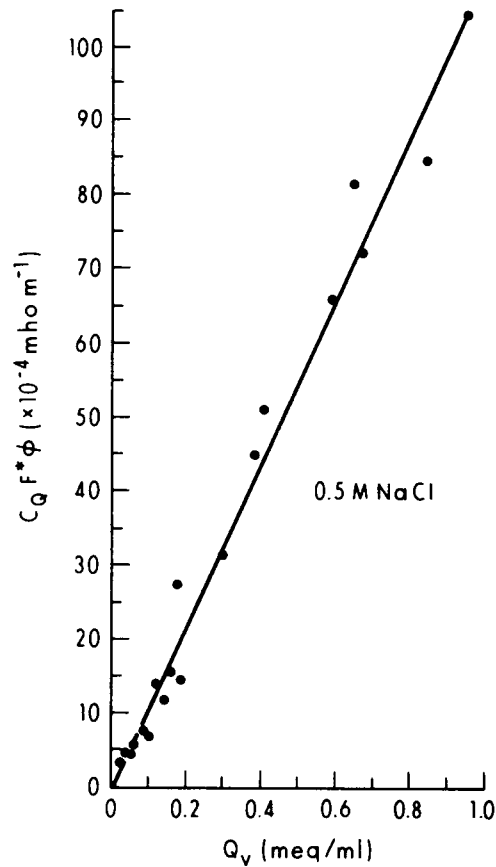


FIG. 8. Regression of $C_Q F^* \phi$ on Q_v at 0.5 M NaCl.

Figure 7 plots $(Q_v)_{ws}$ versus $(Q_v)_{anal}$. There appears to be no systematic difference between analytical determinations of Q_v on the same or from adjacent samples. A linear regression of the form

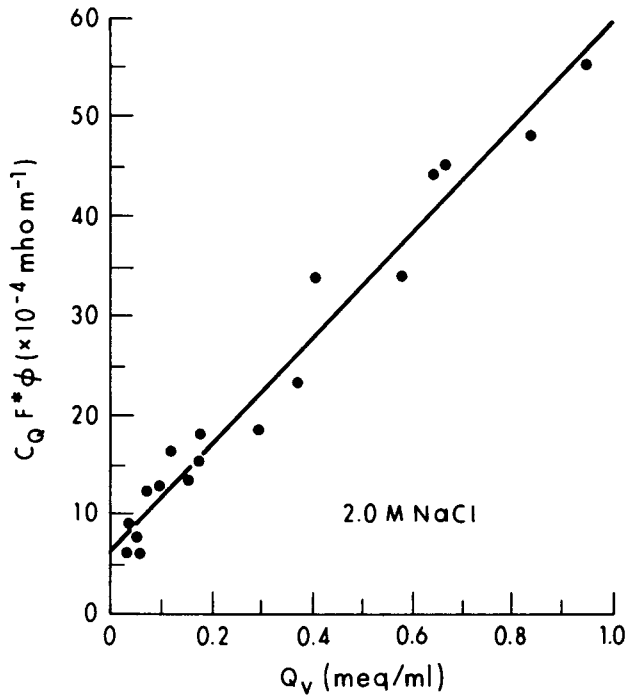
$$(Q_v)_{ws} = a(Q_v)_{anal} + b$$

yields values of $a = 1.031 \pm 0.068$ and $b = -0.009 \pm 0.021$ meq/ml with a correlation coefficient of 0.985.

To test the validity of equation (24), $C_Q F_q = C_Q F^* \phi$ versus Q_v was plotted for each rock at each of five salinities. Figures 8

Table 3. Regression analysis of the quadrature conductivity data. $C_Q F^* \phi = \bar{\lambda} Q_v + b$.

Salinity	Number of Samples	Correlation Coefficient, r	$\bar{\lambda}$ (10^{-4} mho-m ⁻¹ -meq ⁻¹ -ml)	b (10^{-4} mho-m ⁻¹)
2.0 M	18	.983	53.03 + 4.28	6.20 + 1.40
1.0 M	19	.990	91.17 + 5.33	1.25 + 1.70
0.5 M	19	.991	111.79 + 6.35	-.72 + 2.04
0.1 M	19	.980	92.79 + 7.85	.18 + 2.40
0.01 M	15	.964	45.51 + 6.13	.37 + 0.93

FIG. 9. Regression of $C_Q F^* \phi$ on Q_v at 2.0 M NaCl.

and 9 are examples of these plots. For these plots (Q_v)_{anal} was used for the particular sample when available; otherwise (Q_v)_{ws} was substituted. A linear regression, $C_Q F^* \phi = aQ_v + b$, was performed on the data sets for each of the five salt concentrations. The slopes and intercepts of the regression lines are given in Table 3, as well as the correlation coefficients and the 95 percent confidence belts for the regression coefficients.

If the slope of each regression line is interpreted as $\bar{\lambda}$, then it is clear that equations (23b) and (24) accurately describe the data at all salinities except 2.0 M NaCl. The reasons for the positive intercept with 2.0 M NaCl will be discussed in the next subsection.

In order to test the consistency of Q_v values obtained from Waxman-Smiths conductivity data and those obtained from quadrature conductivity, we performed an orthonormal regression of BQ_v from

$$BQ_v = F^*C_l - C_w \quad (34)$$

against $\bar{\lambda}Q_v$, from

$$\bar{\lambda}Q_v = C_Q F^* \phi. \quad (35)$$

Results are tabulated in Table 4; the correlation coefficient is 0.96 or greater for each salinity except 0.01 M NaCl.

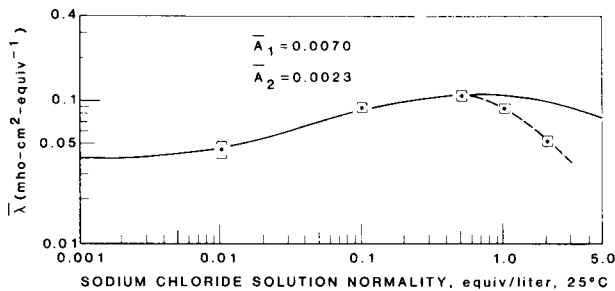
Table 4 also includes values of $\bar{\lambda}/B_{\text{calc}}$ for comparison, where the $\bar{\lambda}$ values were obtained from the slopes given in Table 3 and the B_{calc} values were calculated from equation (12).

Discussion of data

A plot of $\bar{\lambda}$ versus sodium chloride solution normality (taken from Table 3) is given in Figure 10. It is evident that equations (23b) and (24) are a good description of the behavior of quadrature conductivity over a wide range of salinity (0.01 M–1.0 M NaCl). As expected, $\bar{\lambda}$ is only a weak function of salinity.

In order to determine values of the parameters $\bar{A}_1$ and $\bar{A}_2$ in equation (21), the best fit for $\bar{\lambda}$ was obtained using only salinities ≤ 0.5 M NaCl. The solid curve in Figure 10 is obtained with parameters $\bar{A}_1 = 0.0070$ and $\bar{A}_2 = 0.0023$. From these values of $\bar{A}_1$ and $\bar{A}_2$, it appears that the contribution to quadrature conductivity from displacement of clay exchange cations is of the same order of magnitude as that due to displacement of electrolyte solute by membrane blockage.

Although equation (21) can be well fit in the low-salinity range, it deviates from experimental results at concentrations greater than 1.0 M NaCl. The reason is that equation (21)

FIG. 10. Equivalent quadrature conductance $\bar{\lambda}$ versus NaCl solution normality at 25°C.Table 4. Orthonormal regression, $\bar{\lambda}Q_v = a(BQ_v) + b$.

Salinity	Correlation Coefficient, r	$a(\times 10^{-4})^*$	$b(\times 10^{-4} \text{ mho-m}^{-1})$	$\bar{\lambda}/B_{\text{calc}}^\ddagger$
2.0 M	.969	12.887 ± 1.942	7.381 ± 2.401	13.9 ± 1.1
1.0 M	.956	20.632 ± 3.554	1.022 ± 4.965	24.1 ± 1.4
0.5 M	.983	25.929 ± 2.797	1.440 ± 3.549	31.7 ± 1.8
0.1 M	.970	31.754 ± 4.858	-0.135 ± 3.012	47.2 ± 4.0
0.01 M	.817	61.146 ± 29.450	-1.072 ± 2.945	54.6 ± 7.4

* $a = \Delta(\bar{\lambda}Q_v)/\Delta(BQ_v)$

† $\bar{\lambda}$ from slopes, Table 3.

B_{calc} from equation (12).

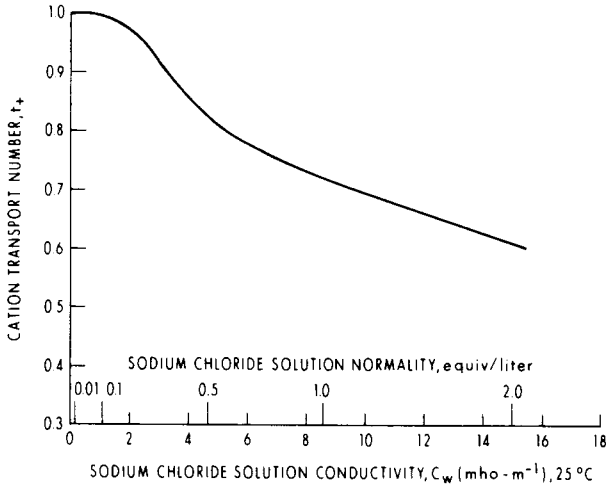


FIG. 11. Variation of cation transport number t_+ with electrolyte conductivity.

assumes a cation transport number t_+ equal to unity through the clay-rich zone, i.e., no anion leak occurs through this zone. In fact, the diffuse portion of the electrical double layer responsible for restrictive anion transport is increasingly compressed at higher salinities (Grahame, 1947); hence the magnitude of the membrane blockage effect and of $\bar{A}_2$ is reduced. At high-salinity levels, the anion leak through pores in clay-rich zones must increase with increasing salinity because the ability of these zones to restrict chloride ion transport is progressively swamped out by higher interstitial chloride ion concentrations. The more general equation (20) must then be used to allow for cation transport numbers less than unity. One approach is to set t_+ equal to unity at low salinities (0.01–0.1 M NaCl) and to determine empirically its dependence at higher salinities. This results in values of the 25°C parameters $\bar{A}_1 = 0.00604$ and $\bar{A}_2 = 0.00397$ and the dependence of t_+ on salinity shown in Figure 11. The decrease of t_+ at high salinities is qualitatively consistent with other laboratory measurements of chloride and sodium ion transport in cation exchange membranes, shaly sands, and shales (e.g., Smits, 1969). The preceding treatment of quadrature conductivity data yields average values for $\bar{\lambda}$ and the $\bar{A}_1$, $\bar{A}_2$, and t_+ parameters.

Alternatively, these data can be treated—albeit less conveniently—as dichotomous, depending upon the relative importance of the clay-exchange conductance and membrane-blockage polarization mechanisms. The quadrature conductivity data in Table 2 indicate that, for “clean” sandstones with $Q_v \leq 0.1$ meq/ml, C_Q ’s and hence individual λ s continue to increase with increasing NaCl concentration from 0.5 to 2.0 M. By contrast, for the shalier samples with $Q_v \geq 0.15$ meq/ml, C_Q values increase to maxima at concentrations of ~ 0.4 – 0.5 M NaCl, but then decrease as concentrations increase to 2.0 M NaCl. These results may be interpreted as follows. For the low Q_v samples, the membrane blockage contribution to the polarization is negligible and the clay counterion conductance term dominates, i.e., $\lambda \approx A_1 B$. For concentrations greater than 0.4 M NaCl, the clay counterion equivalent conductivity B is constant. However, with increased clay contents and higher Q_v ’s, the membrane blockage effect becomes significant and λ is represented by the formalism of equation (20). This interpretation is supported by the regression analysis cited in Table 3. At 2.0 M NaCl (Figure 9), the linear regression $(C_Q F^* \phi) = a Q_v + b$ results in a nonzero and positive b intercept. To a large extent, this is caused by reduction in membrane blockage effect in the shalier samples at high salinities. In addition, part of the positive intercept of $C_Q F^* \phi$ versus Q_v at 2.0 M NaCl (Figure 9) probably is due to residual sample polarization of about 0.1 mrad from sources other than clays. This residual polarization is absent in samples of silicon carbide and clay-free limestone and dolomite. This allows us to exclude both inductive coupling and polarization of electrodes and sample holders as sources of residual polarization (refer to Appendices A and B). These background effects are due to specific material in the sandstones themselves, a response from either quartz sand or from trace amounts of conductive metallic minerals. An extrapolation of laboratory measurements by Collett (1959) yields an estimated phase angle of 0.15 mrad for artificial sand samples containing 0.1 percent by volume pyrite. Metallic minerals in rocks, particularly metallic sulfides such as pyrite, chalcopyrite, and galena, generate a large IP response caused by oxidation-reduction reactions at the metal surfaces activated by applied current (Wait, 1959). Since the presence of pyrite is a source of noise for IP applications in shaly sands, we surveyed the frequency and amounts of pyrite in typical sandstone reservoirs. Of 50 sandstones described in Shell data, pyrite was reported in only eight samples and, of those, only three contained more than 0.5

Table 5. Regression analysis. $C_Q F^* = \bar{\lambda}_a Q_v + b$.

Salinity	Correlation Coefficient, r	$\bar{\lambda}_a (x 10^{-4} \text{ mho-m}^{-1}\text{-meq}^{-1}\text{-ml})$	$b (x 10^{-4} \text{ mho-m}^{-1})$
2.0 M	.951	254.98 ± 36.38	28.54 ± 11.86
1.0 M	.952	436.97 ± 58.77	7.12 ± 18.62
0.5 M	.954	536.19 ± 70.37	-2.17 ± 22.30
0.1 M	.941	390.59 ± 59.15	11.96 ± 15.98
0.01 M	.917	225.12 ± 48.25	3.81 ± 7.34

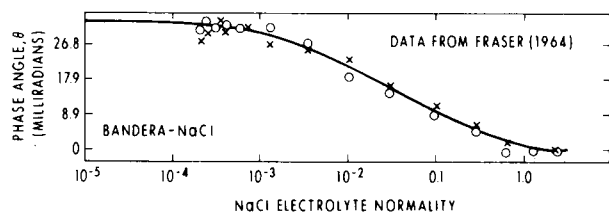


FIG. 12. IP model fit to data of Fraser (1964) for two Bandera sandstone samples.

percent by bulk volume. However, the presence of redox reactions at metal/water interfaces can be differentiated from clay ionic mechanisms by examination of harmonic content (Olhoeft and Scott, 1980; Anderson, 1981; Klein and Sill, 1982). Pyrite concentrations greater than a few percent can also be detected using the Litholog® (photoelectric bulk density tool) because of the high photoelectric cross-section of FeS_2 (Clavier et al., 1976).

Equation (24) represents a good fit to the data (for sands with porosity range of about 12–24 percent) when $F_q = F^*\phi$ [equation (23b)] is used. However, it is still possible that the formation factor for quadrature conductivity F_q is the same as for in-phase conductivity F^* , i.e.,

$$C_Q = \frac{1}{F^*} \lambda_a Q_v. \quad (36)$$

The results of the linear regression $C_Q F^* = aQ_v + b$ are given in Table 5. Although there is more scatter in the $C_Q F^*$ versus Q_v plots than when $F_q = F^*\phi$ is used, nevertheless this represents a good fit for a petrophysical data set. The salinity dependence of λ_a is nearly identical to that of $\tilde{\lambda}$; in fact, $\lambda_a = 4.71\tilde{\lambda}$.

Low electrolyte concentrations

In the limit of low electrolyte concentration ($C_w \ll BQ_v$), $C_w \rightarrow 0$ and θ and θ' in equations (28) and (29) approach

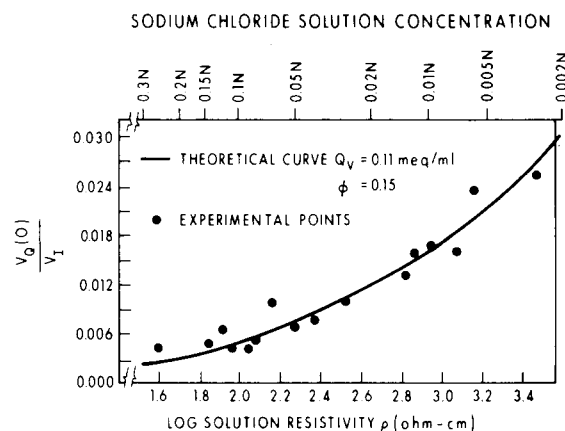


FIG. 13. IP model fit to data of Henkel and Collins (1961).

limiting values

$$\theta, \theta' \rightarrow \frac{(\tilde{\lambda})_{C_w=0}}{\phi B_{C_w=0}}, \quad (37)$$

where $(\tilde{\lambda})_{C_w=0}$ and $B_{C_w=0}$ are the values of $\tilde{\lambda}$ and B for shaly sand in equilibrium with distilled water. The phase angle in equation (37) is independent of both Q_v and S_w in this limit. From Figures 3 and 10 the values of $B_{C_w=0}$ and $(\tilde{\lambda})_{C_w=0}$ are 6.51 and 0.0395 mho-cm²-equiv⁻¹, respectively. When these values are substituted, equation (37) predicts that in all shaly sands the phase angle will approach the universal value in fresh water of

$$\theta_{\max} = \theta'_{\max} = \frac{0.0061}{\phi}. \quad (38)$$

This is about 30 mrad for a rock of 20 percent porosity. (This analysis assumes that $F_q = F^*\phi$.)

Although the lowest salinity used in our study was 0.01 N NaCl, Fraser (1964) reported IP measurements in sandstone cores where salinity was varied from 1.0 N to 10⁻⁴ N NaCl. Figure 12 reproduces his data for two samples of Bandera sandstone. The solid-line curve is our fit of equation (28) to his data using $Q_v = 0.35$ meq/ml and $\phi = 0.19$, reasonable values for Bandera sandstone. All of Fraser's samples (Berea, Boise, and Bandera sandstones) reached maximum polarizations of 27–30 mrad of phase at 10⁻⁴ N NaCl. This agrees with the prediction of equation (38) in the limit of distilled water as the pore electrolyte.

Henkel and Collins (1961) also published IP data extending to very low electrolyte concentrations. Figure 13 shows their data for the chargeability of a sandstone core as a function of salinity; the solid-line curve is a fit of equation (28) where $Q_v = 0.11$ meq/ml and $\phi = 0.15$ were assumed.

Temperature dependence

The temperature dependence of in-phase and quadrature conductivities of two sandstones was measured between -5° and 95°C (Table 6). These samples, Berea V107 and Big Foot 3306F, represent the extremes of a "clean" and a very shaly

Table 6. Variation of conductance with temperature.

T (°C)	C _I (mho-m ⁻¹)	C _Q (10 ⁻⁴ mho-m ⁻¹)	C _Q (T)/C _Q (25°C)
BEREA V107; 0.01 N NaCl, 25°C Q _v = 0.06 meq/ml; φ = 0.169			
25	.0207	1.83	1.00
50	.0320	3.02	1.65
75	.0424	4.26	2.33
100	.0531	5.60	3.06
BIG FOOT 3306F; 0.46 N NaCl (from C _I versus C _w data), 25°C Q _v = 0.95 meq/ml; φ = 0.171			
25	.170	15.0	1.00
50	.304	26.1	1.74
75	.469	39.0	2.59
100	.585	47.8	3.18

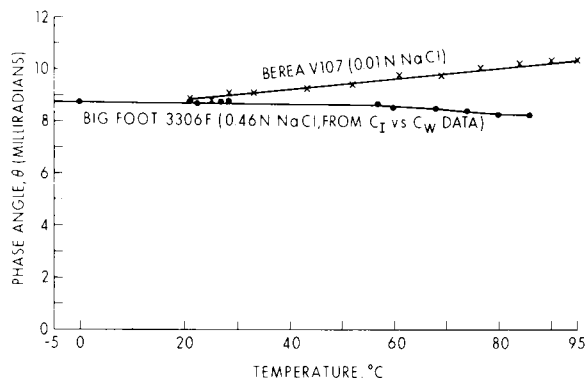


FIG. 14. Variation of phase angle with temperature.

sandstone. The Berea sand was equilibrated with 0.01 N NaCl at 25°C; the Big Foot sand was equilibrated with 0.46 N NaCl at 25°C, as estimated from C_I versus C_w data. For these samples, as shown in Figure 21, the phase angle is practically independent of temperature. The temperature independence of the phase angle in shaly sands has also been reported by Kern et al. (1977) and by Keevil and Ward (1962).

In fact, a small increase in the phase angle was observed for the Berea sample, from 8.84 to 10.55 mrad, as temperature increased from 25° to 100°C. Conversely, over the same temperature interval, the phase angle for the shaly Big Foot sample was reduced from 8.82 to 8.17 mrad.

These trends are explained in terms of the two polarization mechanisms: clay exchange-cation displacement and membrane blockage of electrolyte anions. As described previously, clay cation displacement is the dominant polarization effect for "clean" sandstones. If $C_w \gg BQ_v$ for the low- Q_v rock, the phase angle for a given sample is essentially proportional to the ratio B/Λ [equations (15) and (28)]. Conversely, for an extremely shaly, high- Q_v rock, the membrane blockage effect dominates with $BQ_v \geq C_w$. The phase angle for this rock will approach proportionally to Λ/B [equations (16) and (28)]. Waxman and Thomas (1974) observed that the equivalent conductance of the clay exchange cations B increases more rapidly with increasing temperature than the equivalent conductivity of the electrolyte Λ . Therefore, with increasing temperature, some increase or decrease in phase angle is expected for the limiting cases of clean or shaly sands, respectively.

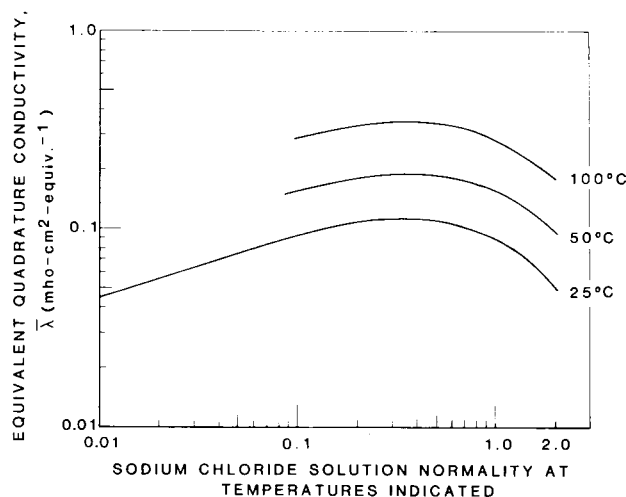
Values of $\bar{\lambda}$ as a function of sodium chloride concentration at 50° and 100°C are given in Figure 15. The calculation, based on the limited data of Table 6, is described in Appendix C.

IP MEASUREMENTS OF OIL SATURATION

For oil-bearing shaly sands ($S_w < 1$), equation (27) can be written as

$$C_Q = \frac{S_w^{n^*}}{F^* \phi} \bar{\lambda} \frac{Q_v}{S_w} = \frac{S_w^p}{F^* \phi} \bar{\lambda} Q_v, \quad (39)$$

where n^* is the saturation exponent in the Waxman-Smiths equation (13) and $p = n^* - 1$ is the quadrature saturation exponent. Note that, because $n^* \approx 2$ in most instances, the model

FIG. 15. Variation of $\bar{\lambda}$ with temperature.

predicts that quadrature conductivities in oil-bearing sands are reduced by the factor S_w from their values at 100 percent water-saturated conditions.

The dependence of C_Q on oil saturation as described by equation (39) is supported by two previous studies of IP in shaly sands. Sill (1964) measured the effects of oil saturation on IP in three cores having $Q_v \leq 0.7$ meq/ml. He concluded that the increase in percent frequency effect was proportional to the increase in "effective b -factor," i.e., the modified b -factor of Hill and Milburn 1956 in partially oil-saturated sands, obtained by replacing b with b/S_w . Since the Hill-Milburn b -factor in water-saturated sands is approximately equal to $-Q_v/7$, this is a verification of the assumption described by equation (26) and incorporated in equations (29) and (39).

More recently, Hoyer and Rumble (1976) and Kern et al. (1977) reported that the low-frequency effective dielectric constant K_{eff} varied with water saturation as

$$K_{eff} = K_{100} S_w^p, \quad (40)$$

where K_{100} is the effective dielectric constant at 100 percent water saturation and $p = 1.00 \pm 0.20$. Data for three cores were presented where the range of water saturation was $0.3 \leq S_w \leq 1.0$. Since the low-frequency dielectric constant is related to C_Q by $K_{eff} = C_Q/\omega\epsilon_0$ [equation (7)], these data also confirm the quadrature saturation dependence in equation (39), based on the assumption that $n^* \approx 2$.

While the results of both Hoyer and Sill are consistent with equation (39), neither experiment was conducted at capillary equilibrium oil saturation. In addition, in-phase conductivity data were not reported, so the saturation exponents n^* are not available to verify the relation

$$p = n^* - 1. \quad (41)$$

The present study was undertaken to examine the dependence of C_Q on oil saturation at capillary equilibrium conditions. Plugs from four formations with a wide range of shaliness were used to measure in-phase and quadrature conductivities, so that the saturation exponents n^* and p could be determined independently.

Table 7. Petrophysical properties of samples.

Sample No.	Well	Age	Formation	Depth Ft.	Porosity	Grain Density, g/cc	Q_v meq/ml
V104	-	-	Berea Outcrop	-	0.188	2.659	0.06
V106	-	-	Berea Outcrop	-	0.188	2.659	0.066
V110	-	-	Berea Outcrop	-	0.189	2.662	0.06
V112	-	-	Berea Outcrop	-	0.188	2.659	0.06
1883	Shell Plow Realty, Texas	Lower Eocene	Lower Wilcox	~10,825	0.133	2.646	0.120
3371	Middle Ground Shoal, Alaska	Oligocene	Kenai	~8,259	0.142	2.645	0.321
3372C	Middle Ground Shoal, Alaska	Oligocene	Kenai	~8,259	0.127	2.637	0.349
3374	Middle Ground Shoal, Alaska	Oligocene	Kenai	~8,259	0.146	2.649	0.236
3307C	Unit No. 169, Frio County, Texas	Cretaceous	Big Foot	~3,580	0.118	2.698	1.25

Experimental procedures and data presentation

Petrophysical and geologic descriptions of the various sandstones are given in Table 7.

The measurement cells and experimental procedures related to the establishment of various oil saturations in the sand plugs are described in Appendices A and B.

A sample was assumed to be in capillary equilibrium at a given oil saturation when the electrical resistivity and the oil volume introduced into the plug were constant within experi-

mental error over a 24-hour period. The measurement of each saturation curve required 4–8 weeks. Only limited data ($S_w \cong 0.85$ to 0.90) could be obtained for two of the more shaly samples (3374, 3307C) due to the high oil pressures required to attain lower water saturations and the restriction of the oil displacement pressure (<20 psi) of the capillary diaphragms. After other measurements were completed, Q_v determinations were made on crushed samples as described previously. Oil saturations were determined from burette readings and from petrophysical determinations of pore volumes; the estimated error is $\Delta S_w/S_w = \pm 2$ percent.

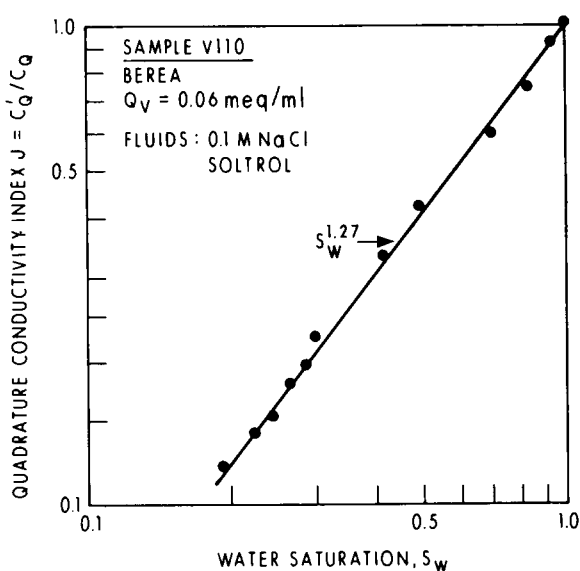


FIG. 16. Quadrature conductivity index versus water saturation, sample V110.

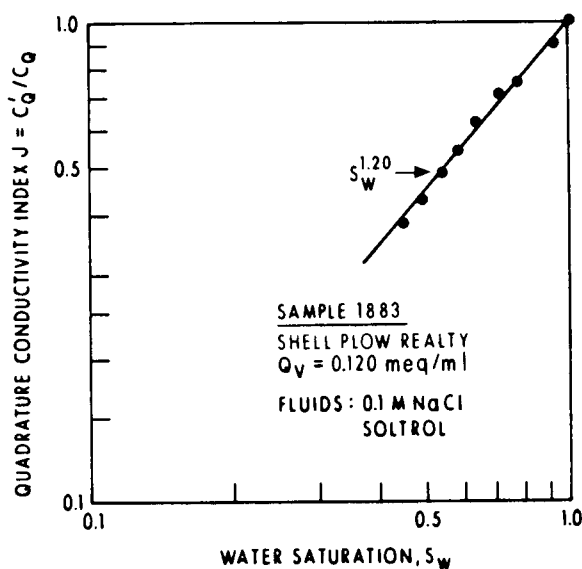


FIG. 17. Quadrature conductivity index versus water saturation, sample 1883.

Table 8 includes the brine salinity and type of oil used for each plug. Most of the samples were saturated with 0.1 N NaCl solution, except Berea V112 (0.01 N NaCl) and Big Foot 3307C (1.0 N NaCl). Soltrol was used as the nonwetting phase except in Berea samples V104 and V106, where El Dorado and Ventura crude oils were substituted.

Discussion of data

By analogy with the resistivity index I ($I = R_i/R_o = C_i/C'_i$), we define the quadrature conductivity index J as the ratio of C'_Q for $S_w < 1$ to C_Q for $S_w = 1$. Complex conductivity data taken at 30 Hz are reported in Table 9 for fully brine-saturated ($S_w = 1$) sands as well as the values of I and J as a function of S_w . Figures 16 and 17 are typical examples of J versus S_w plotted on log-log coordinates. The solid lines are the least-squares regressions of the function $J = aS_w^p$ on S_w .

Table 8 lists the number of data points for each sample, the coefficients a and quadrature saturation exponents p , correlation coefficients r , and standard errors of estimate for a and p . For the sample set, $p = 1.11 \pm 0.17$.

The consistency between $p \equiv n^* - 1$ predicted by equation (39) and the in-phase saturation exponent n^* in the Waxman-Smiths equation was checked by computing n^* from the in-phase conductivity data. The known values of C_w and Q_v were used to solve the Waxman-Smiths equation for I_{calc} at each equilibrium saturation. The value of n^* was determined by minimizing the sum of the squares of the deviations of I_{calc} from I_{meas} .

Table 10 compares n^* computed in this way with $(p + 1)$ obtained from the quadrature conductivity data. For the sample set, the mean value of $[n^* - (p + 1)] = -0.047$ with a standard deviation of ± 0.204 .

These results indicate that the saturation dependence $S_w^{(n^*-1)} = S_w^p$ predicted for C'_Q in equation (39) is in good agreement with the Waxman-Smiths saturation exponent n^* . The mean value of p for our data set agrees with that reported by Hoyer and coworkers under nonequilibrium conditions. Within experimental error, the quadrature saturation exponents show no dependence on salinity or type of oil.

CONCLUSIONS

A model and equations have been developed to account for the effects of shaliness, brine conductivity, and oil saturation on the reactive part of the electrical conductivity of shaly sands. These equations use the same petrophysical variables as the Waxman-Smiths in-phase conductivity equation.

Several aspects of the theory have been verified experimentally. For water-bearing sands, the quadrature conductivity C_Q is directly proportional to the shaliness parameter Q_v and only weakly dependent upon electrolyte concentration. Empirical values of $\bar{\lambda}$, the equivalent quadrature conductivity, were determined as a function of sodium chloride concentration.

Both quadrature conductivities and phase angles are practically independent of frequency. Phase angles are approximately independent of temperature.

Table 8. Quadrature saturation exponents and associated error analysis.

Sample	Salinity	Oil Type	No. of Data Points	Correlation Coefficient, r	p	Standard Error of Estimate, S_p	a	Standard Error of Estimate, S_a
V104 Berea	0.1 N	Eldorado Crude	6	0.99	1.180	0.0237	0.994	0.016
V106 Berea	0.1 N	Ventura Crude	8	0.99	1.284	0.0439	0.980	0.013
V110 Berea	0.1 N	Soltrol	14	0.99	1.274	0.0242	0.981	0.025
V112 Berea	0.01 N	Soltrol	7	0.99	0.777	0.012	1.006	0.012
1883 Shell Plow Realty	0.1 N	Soltrol	9	0.98	1.202	0.023	1.017	0.012
3371 Middle Ground Shoal	0.1 N	Soltrol	7	0.99	1.170	0.028	0.993	0.011
3372C Middle Ground Shoal	0.1 N	Soltrol	6	0.99	1.159	0.021	1.002	0.006
3374 Middle Ground Shoal	0.1 N	Soltrol	2	1.00	0.93	-	1.00	-
3307C Big Foot	1.0 N	Soltrol	2	1.00	1.03	-	1.00	-

Table 9.1. In-phase and quadrature conductivities for brine-saturated sands; in-phase resistivity and quadrature conductivity indices, water saturations for oil-bearing sands.

Sample Number	S_w	I	J (at 30 Hz)
V104	1.000	1.000	1.000
$C_I = 0.0601 \text{ mho-m}^{-1}$	0.630	2.895	0.565
$C_Q = 0.000118 \text{ mho-m}^{-1}$	0.601	3.167	0.543
	0.489	4.295	0.440
	0.436	5.313	0.373
	0.412	5.865	0.347
V106	1.000	1.000	1.000
$C_I = 0.0561 \text{ mho-m}^{-1}$	0.958	1.126	0.945
$C_Q = 0.000142 \text{ mho-m}^{-1}$	0.917	1.356	0.872
	0.804	1.871	0.718
	0.792	1.940	0.710
	0.662	2.697	0.574
	0.636	2.768	0.557
	0.614	2.897	0.537

Table 9.2. In-phase and quadrature conductivities for brine-saturated sands; in-phase resistivity and quadrature conductivity indices, water saturations for oil-bearing sands.

Sample Number	S_w	I	J (at 30 Hz)
V110	1.000	1.000	1.000
$C_I = 0.0656 \text{ mho-m}^{-1}$	0.945	1.196	0.911
$C_Q = 0.000154 \text{ mho-m}^{-1}$	0.836	1.741	0.738
	0.701	2.374	0.592
	0.494	3.932	0.414
	0.420	5.279	0.327
	0.364	6.658	0.295
	0.333	7.987	0.259
	0.302	9.612	0.224
	0.287	11.259	0.196
	0.266	12.838	0.177
	0.243	15.714	0.154
	0.224	17.835	0.140
	0.196	21.391	0.120
V112	1.000	1.000	1.000
$C_I = 0.0105 \text{ mho-m}^{-1}$	0.878	1.187	0.906
$C_Q = 0.000103 \text{ mho-m}^{-1}$	0.472	2.458	0.570
	0.426	2.755	0.525
	0.381	3.105	0.474
	0.345	3.562	0.442
	0.300	4.066	0.387

Table 9.3. In-phase and quadrature conductivities for brine-saturated sands; in-phase resistivity and quadrature conductivity indices, water saturations for oil-bearing sands.

Sample Number	S_w	I	J (at 30 Hz)
1883	1.000	1.000	1.000
$C_I = 0.0282 \text{ mho-m}^{-1}$	0.913	1.160	0.903
$C_Q = 0.000118 \text{ mho-m}^{-1}$	0.778	1.490	0.754
	0.721	1.680	0.699
	0.644	1.980	0.614
	0.586	2.320	0.543
	0.542	2.650	0.484
	0.489	3.030	0.431
	0.451	3.490	0.381
3371	1.000	1.000	1.000
$C_I = 0.0467 \text{ mho-m}^{-1}$	0.848	1.468	0.804
$C_Q = 0.000363 \text{ mho-m}^{-1}$	0.755	1.755	0.722
	0.697	2.047	0.656
	0.632	2.522	0.571
	0.592	2.738	0.544
	0.570	2.994	0.513
3372C	1.000	1.000	1.000
$C_I = 0.0407 \text{ mho-m}^{-1}$	0.977	1.105	0.976
$C_Q = 0.000335 \text{ mho-m}^{-1}$	0.826	1.502	0.797
	0.759	1.707	0.737
	0.679	2.079	0.643
	0.639	2.331	0.591

Table 9.4. In-phase and quadrature conductivities for brine-saturated sands; in-phase resistivity and quadrature conductivity indices, water saturations for oil-bearing sands.

Sample Number	S_w	I	J (at 30 Hz)
3374	1.000	1.000	1.000
$C_I = 0.0565 \text{ mho-m}^{-1}$	0.895	1.193	0.902
$C_Q = 0.000489 \text{ mho-m}^{-1}$			
3307C	1.000	1.000	1.000
$C_I = 0.208 \text{ mho-m}^{-1}$	0.852	1.293	0.848
$C_Q = 0.000790 \text{ mho-m}^{-1}$			

Table 10. Comparison of $(p + 1)$ derived from IP measurements with n^* from in-phase conductivity measurements.

Sample	R_w , ohm-m	$B^\dagger$	Q_v , Equiv/Liter	$R_w B Q_v$	n^*	$(p + 1)$
V104	0.937	2.033	0.06	0.114	2.17	2.18
V106	0.937	2.033	0.066	0.126	2.43	2.28
V110	0.937	2.033	0.06	0.114	2.11	2.27
V112	8.44	0.942	0.06	0.477	1.63	1.78
1883	0.937	2.033	0.120	0.229	1.81	2.20
3371	0.937	2.033	0.321	0.611	2.40	2.17
3372C	0.937	2.033	0.349	0.665	2.35	2.16
3374	0.937	2.033	0.236	0.450	1.79	1.93
3307C	0.094	3.82	1.25	0.458	1.89	2.03
				Average	2.06	2.11

† Units of B; liter equiv⁻¹ohm⁻¹m⁻¹

For oil-bearing sands, the IP model indicates that quadrature conductivities should be reduced from their values at $S_w = 1$ by the factor $S_w^{n^*-1}$ or S_w^p , where n^* is the in-phase saturation exponent and p is the quadrature saturation exponent. The relation $p = n^* - 1$ was confirmed by independent experimental determinations of n^* and p at capillary equilibrium.

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APPENDIX A EXPERIMENTAL IP SYSTEM

This section describes the measurement system, sample holders, electrodes, and flow apparatus.

Measurement system

The frequency-domain measurements reported here were

made with a four-electrode bridge in order to eliminate electrode polarization problems inherent in two-electrode systems at low frequencies. As shown in Figure A-1, a sinusoidal current is applied to a series circuit consisting of the sample and a variable RC network. The voltage across the sample and the RC network is equal when the impedance of the RC network is

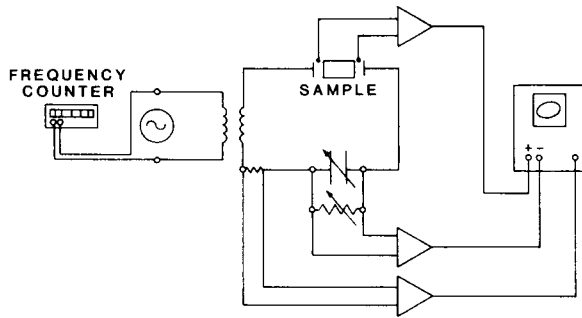


FIG. A-1. Bridge circuit for measuring complex resistivity in frequency domain.

adjusted to equal the sample impedance. The Lissajous pattern on the oscilloscope is used for phase comparison and null detection: at resistive null the oscilloscope trace is an ellipse with horizontal major axis; at both resistive and capacitive balance the ellipse becomes a straight line. Bridge accuracy is 0.01 percent from 1 Hz to 1 KHz.

FET input instrumentation amplifiers are used to isolate the current and potential circuits. The very high input impedance ($10^{13} \Omega$) prevents current flow through the potential circuit, which would polarize the voltage electrodes.

Transformer coupling of the oscillator to the bridge circuit guarantees that no dc bias is applied to the sample and pro-

vides balanced drive for shielding the sample and RC boxes. This minimizes stray coupling at the higher frequencies.

Conductivity measurements were made in the frequency range of 3 Hz to 1 KHz. Inductive coupling limits the upper frequency range to about 1 KHz, without correction.

Sample holders

The basic sample holder used in these experiments is shown in Figure A-2. The sample holder is constructed entirely of Plexiglas® to prevent metallic polarization. Electrical contact with the sample is made through the electrolyte solution in the endpieces. The solution is maintained at the same concentration as the solution saturating the core. While inside the sample holder, the rock can be flushed with brine up to 500 psi internal pressure.

The voltage electrodes are reversible Ag-AgCl, made by chloridizing annealed silver rods using standard electroplating techniques.

The samples are sealed first with a thick epoxy butter to prevent imbibition, then cemented into roughened Plexiglas cylinders with a less-viscous epoxy. The ends of the samples are faced flat with a carbide-tipped cutting tool; no cutting fluid is used.

For the experiments at capillary equilibrium oil saturation, the oil was introduced through a hole drilled into the side of the Plexiglas sleeve encapsulating the sample. Capillary diaphragms through which the water phase was displaced were

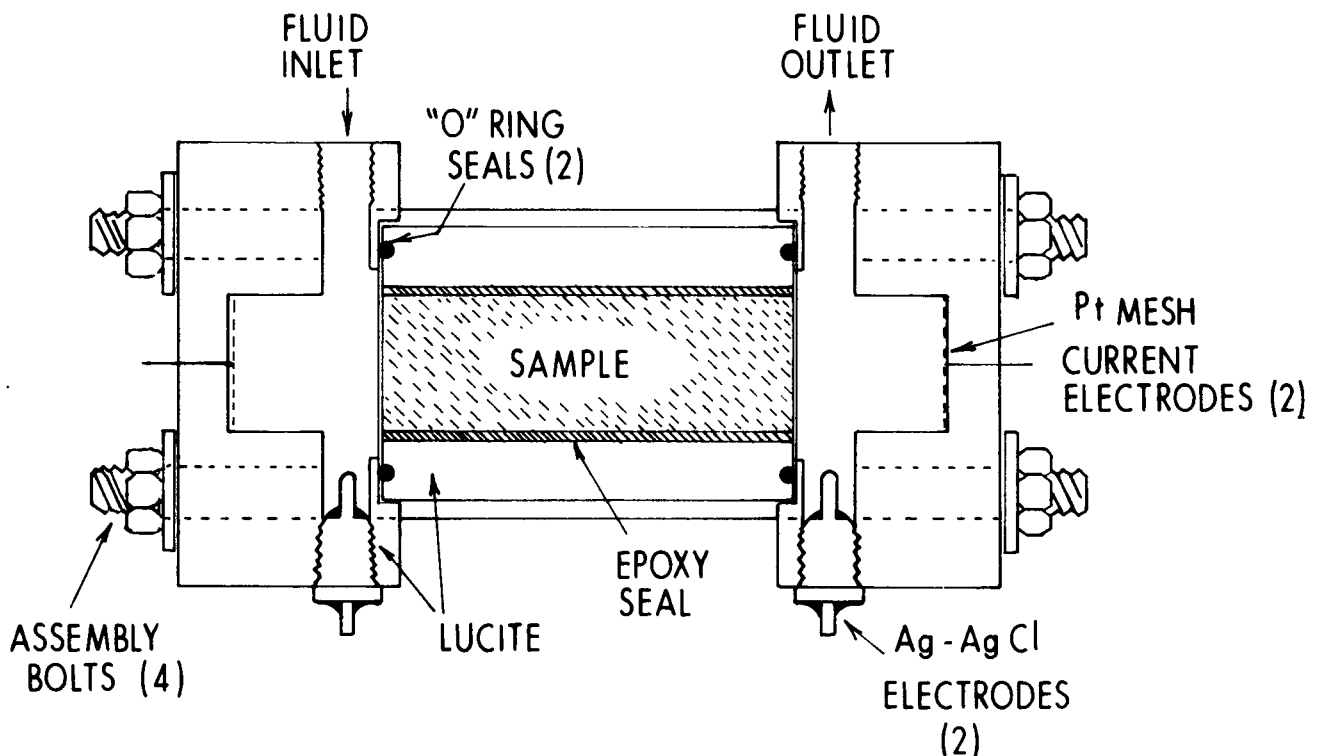


FIG. A-2. Core sample holder for IP measurements.

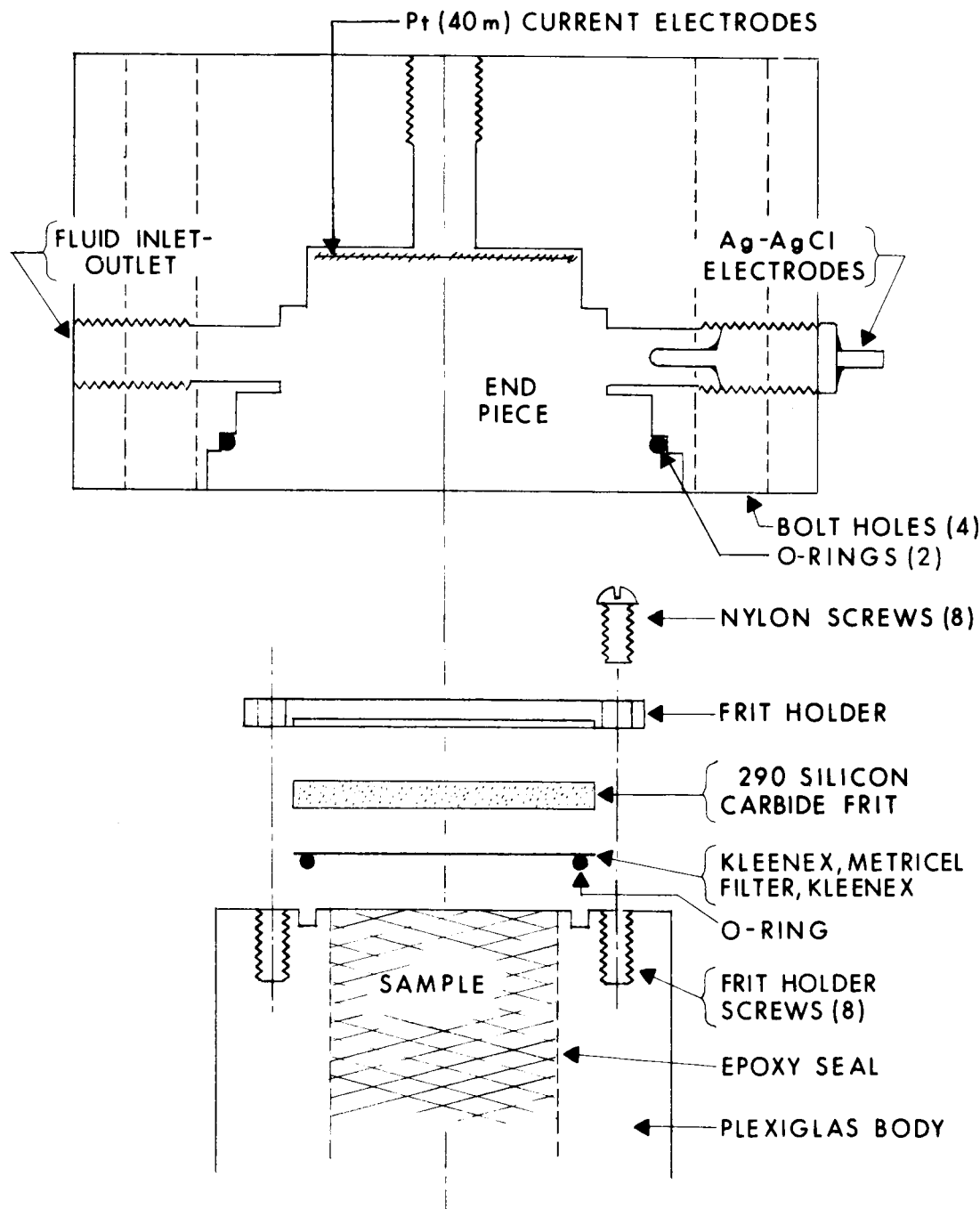


FIG. A-3. Sample and holder, exploded view.

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mounted on each end of the core (see Figure A-3). The capillary diaphragms are brine-saturated Alpha Metrical® filters (Alpha-8, 0.2 micron pore size) with an oil displacement pressure of about 15–20 psi when wet. These thin diaphragms introduce negligible polarization; when wet, the Metrical filters swell and improve core contact. Kleenex® discs are added to ensure capillary continuity. The filters are squeezed against O-rings and sample by a 3/32-inch thick silicon carbide frit backed by a Lucite® ring. Polishing the rock faces and seating the filter

firmly are necessary to prevent an oil layer from developing across the face of the sample.

Corrections are made to the raw data for the silicon carbide frits and capillary diaphragms. As determined in separate experiments, these frits have a series resistance but no polarizability.

The entire sample holder was immersed in a temperature-regulated oil bath in order to take measurements at different temperatures.

APPENDIX B ACCURACY AND VALIDITY CHECKS

Spurious polarization

It is desirable to test a porous but electrically inert sample to verify that the sample holder and electrodes do not generate spurious polarization. Cores of porous silicon carbide, and limestone and dolomite cores free of clay, have been found suitable for this purpose. They exhibit no detectable polarization over the measured frequency range.

As another check, the IP of a uniform sample of Berea sandstone was measured. The sample was then sliced in half and each half remeasured. The phase measurement was unchanged, verifying that a bulk property of the core is measured and sample edge effects are not significant.

Reciprocity and nonpolarizing electrodes

Erroneous results are obtained if there is polarizable material in the current path between the voltage electrodes or if the voltage electrodes are themselves polarizable. This was studied in a series of ten experiments with interchangeable platinum mesh (polarizable) and Ag-AgCl electrodes (nonpolarizable). The results are summarized in Figure B-1 where the current and voltage connections are designated I and V , respectively.

Note that because the rock-plus-electrodes material behaves as a linear system, the function of current and voltage electrodes can be reversed without changing the measured values. This consequence of reciprocity for linear systems is shown by comparing the left and right sides of Figure B-1.

Linearity

The oscilloscope display of bridge balance gives visual indication of deviations from linearity; nonlinear behavior appears as deviation from a simple ellipse or straight line. Current densities were maintained below 10^{-7} A/cm² to ensure linear behavior.

The electrical behavior of silicon carbide plugs and clay-free limestone and dolomite was observed to be linear from 1 Hz to 10 KHz. Small nonlinearities (~ 0.1 – 0.5 percent THD) were observed in shaly sands under certain conditions. The amount of nonlinearity increased with shaliness, suggesting that the nonlinear mechanism is associated with interactions at the clay surfaces. The nonlinearity also increased with decreasing fre-

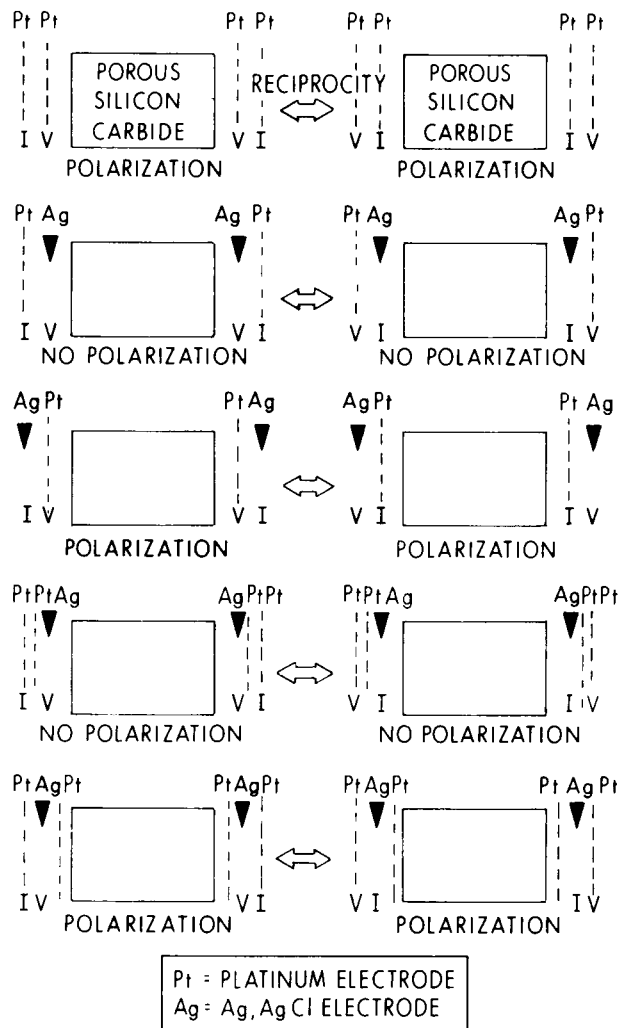


FIG. B-1. Summary of experiments on electrode polarization.

quency below 10 Hz, increasing current density, and increasing oil saturation.

APPENDIX C CALCULATION OF TEMPERATURE DEPENDENCE OF $\bar{\lambda}$

Average values of the ratio $(C_Q)_T/(C_Q)_{25}$ and hence $(\bar{\lambda})_T/(\bar{\lambda})_{25}$ are equal to 1.70 and 3.12 at 50° and 100°C, respectively (Table 6). From these ratios and the $(\bar{\lambda})_{25}$ values at 0.1 and 0.5 N NaCl, the 50° and 100°C $\bar{\lambda}$ values for 0.1 and 0.5 N NaCl were obtained. These were used in equation (20) to solve for $\bar{A}_1$ and $\bar{A}_2$ at both 50° and 100°C. For the two temperatures, values of $\bar{A}_1$ and $\bar{A}_2$ are identical: $\bar{A}_1 = 0.0032$ and $\bar{A}_2 = 0.0058$. These were used in equation (20) to calculate $\bar{\lambda}$ at 50° and 100°C at higher salt concentrations.

Other assumptions and required data were obtained as follows. The cation transport numbers t_+ (Figure 11) were assumed to be temperature independent. Slightly revised values of the clay counterion equivalent conductance B at 50° and 100°C were obtained from Waxman-Thomas data (1974). $B_{\max}(50^\circ\text{C})$ and $B_{\max}(100^\circ\text{C})$ are 94 ± 12 and 166 ± 11 mho-cm²-eq⁻¹, respectively. The equivalent conductivities of sodium chloride solutions as a function of salt concentration were taken from the literature (Chambers, 1958; Noyes, 1908; Washburn, 1929).